

Logic Circuits

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Agenda

- ◆ Logic gates
- ◆ Boolean Algebra
- ◆ NAND and NOR Circuits
- ◆ Logic Minimisation

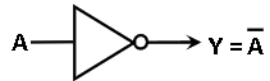
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Logic Gates

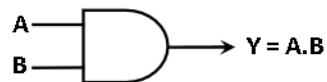
NOT gate (Inverter)



Truth Table of NOT gate

A(I/P)	Y(O/P)
0	1
1	0

AND gate



Truth Table of AND gate

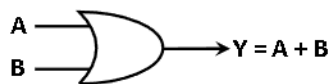
A	B	Y(O/P)
0	0	0
0	1	0
1	0	0
1	1	1

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Logic Gates

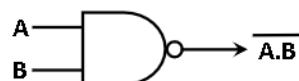
OR gate



Truth Table of OR gate

A	B	Y(O/P)
0	0	0
0	1	1
1	0	1
1	1	1

NAND gate



Truth Table of NAND gate

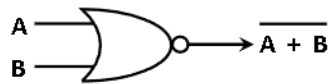
A	B	Y(O/P)
0	0	1
0	1	1
1	0	1
1	1	0

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Logic Gates

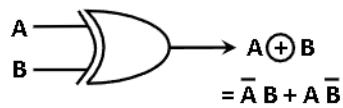
NOR gate



Truth Table of NOR gate

A	B	Y(O/P)
0	0	1
0	1	0
1	0	0
1	1	0

XOR gate



Truth Table of XOR gate

A	B	Y(O/P)
0	0	0
0	1	1
1	0	1
1	1	0

...

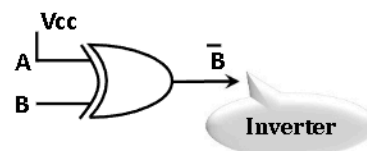
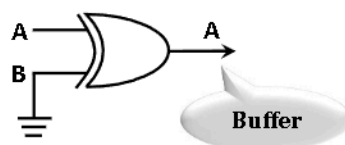
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Properties of XOR Gate

A	B	EXOR
0	0	0
0	1	1
1	0	1
1	1	0

- ✓ if one of the inputs is '0', then the output is same as the second input.
- ✓ If one of the inputs is '1', then, the output is the complement of the second input.



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Properties of XOR Gate

A	B	EXOR
0	0	0
0	1	1
1	0	1
1	1	0

- ✓ Odd parity detector.
- ✓ Even parity generator.

Since, the total no. of 1's in inputs and output is even. So, we can consider it as Even Parity Generator.

Since, No. of 1's in inputs, is odd. Output has become 1. So, we can consider it as odd parity detector.

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Logic Gates

EXNOR gate



Can be called
as Equality
Gate

Truth Table of EXNOR gate

A	B	Y(O/P)
0	0	1
0	1	0
1	0	0
1	1	1

Output is complement of EXOR gate.

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Boolean Algebra

$$(T1) \quad X + 0 = X$$

$$(T2) \quad X + 1 = 1$$

$$(T3) \quad X + X = X$$

$$(T4) \quad (X')' = X$$

$$(T5) \quad X + X' = 1$$

$$(T1') \quad X \cdot 1 = X$$

$$(T2') \quad X \cdot 0 = 0$$

$$(T3') \quad X \cdot X = X$$

$$(T5') \quad X \cdot X' = 0$$

$$(T6) \quad X + Y = Y + X$$

$$(T7) \quad X + (Y + Z) = (X + Y) + Z$$

$$(T8) \quad X \cdot Y + X \cdot Z = X \cdot (Y + Z)$$

$$(T9) \quad X + X \cdot Y = X$$

$$(T10) \quad X \cdot Y + X \cdot Y' = X$$

$$(T6') \quad X \cdot Y = Y \cdot X;$$

$$(T7') \quad X \cdot (Y \cdot Z) = (X \cdot Y) \cdot Z$$

$$(T8') \quad (X + Y) \cdot (X + Z) = X + (Y \cdot Z)$$

$$(T9') \quad X \cdot (X + Y) = X$$

$$(T10') \quad (X + Y) \cdot (X + Y') = X$$

$$(T11) \quad X \cdot Y + Y \cdot Z + X' \cdot Z = X \cdot Y + X' \cdot Z$$

$$(T11') \quad (X + Y) \cdot (Y + Z) \cdot (X' + Z) = (X + Y) \cdot (X' + Z)$$

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Boolean Algebra

Expression

Dual of the Expression

$$A + B = B + A$$

$$A + AB = A$$

$$A + \bar{A}B = A + B$$

$$A \cdot B = B \cdot A$$

$$A \cdot (A + B) = A$$

$$A \cdot (\bar{A} + B) = AB$$

Famous De Morgan Rules

Dual of the Expression

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

Consensus Theorem

Dual of the Expression

$$AB + \bar{A}C + BC = AB + \bar{A}C$$

$$(A + B)(\bar{A} + C)(B + C) = (A + B)(\bar{A} + C)$$

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Proof for Consensus Theorem

$$AB + \overline{A}C + BC = AB + \overline{A}C$$

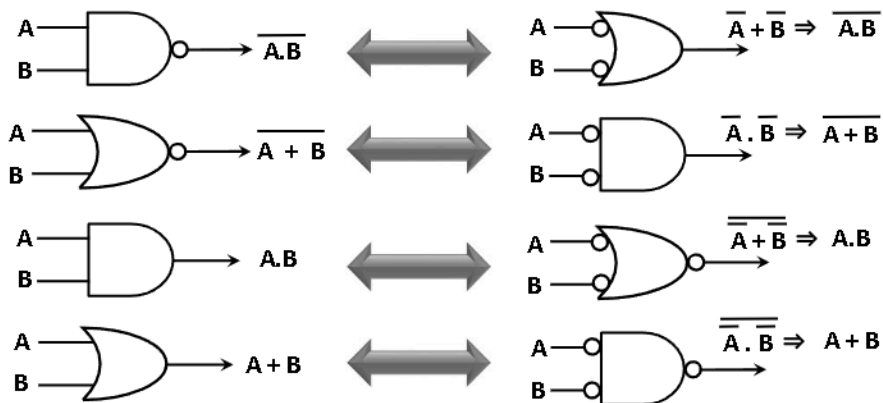
$$\begin{aligned} & AB + \overline{A}C + BC \\ &= AB + \overline{A}C + (A + \overline{A})BC \\ &= AB + \overline{A}C + ABC + \overline{A}BC \\ &= AB(1 + C) + \overline{A}C(1 + B) \\ &= AB + \overline{A}C \end{aligned}$$

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Gate Equivalents

De Morgan's rules.



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Complement

- ♦ The complement of a function F is denoted by F' or $\bar{F}$
- ♦ DeMorgan's theorems to compute the complement of a function.

Another way of finding Complement function

$$F(A,B,C) = A\bar{B} + B\bar{D} + \bar{A}\bar{D}$$

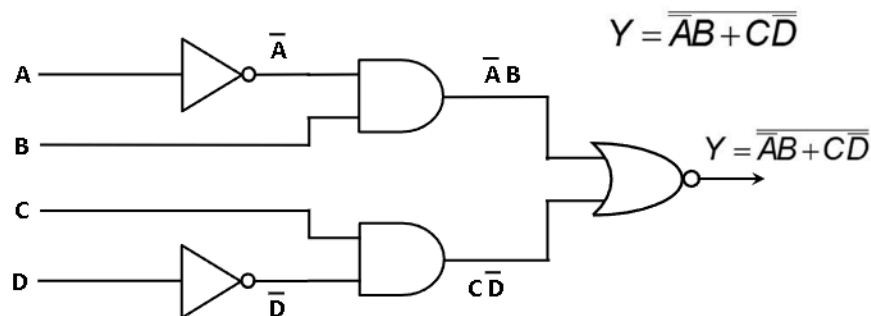
$$\begin{aligned} F(A,B,C) &= (A\bar{B}) + (B\bar{D}) + (\bar{A}\bar{D}) \\ \text{Dual} = F_d &= (A + \bar{B})(B + \bar{D})(\bar{A} + \bar{D}) \\ \text{Comp} = \bar{F} &= (\bar{A} + \bar{B})(\bar{B} + \bar{D})(\bar{A} + \bar{D}) \\ &= \bar{F} = (\bar{A} + B)(\bar{B} + D)(A + D) \end{aligned}$$

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Logic Circuit from function

Implement the function using Logic Gates.



✓ Consider each sum or product term individually and try to obtain them separately by using appropriate logic gates.

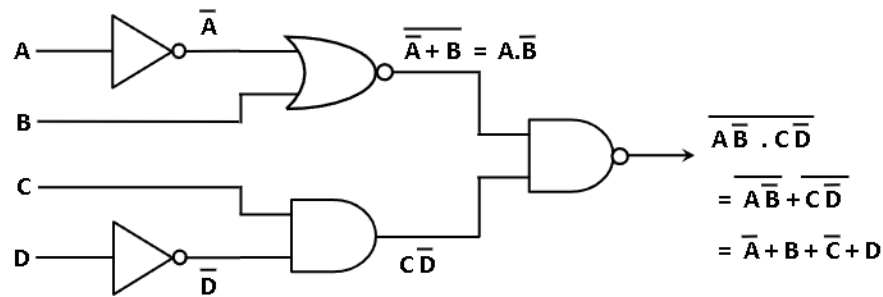
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Function from circuit

Determine the output function from the given circuit.



- ✓ Work progressively from the inputs to the outputs.
- ✓ Derive the logic expressions for the outputs of all the gates.

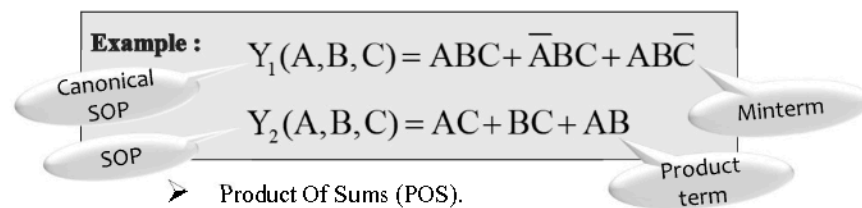
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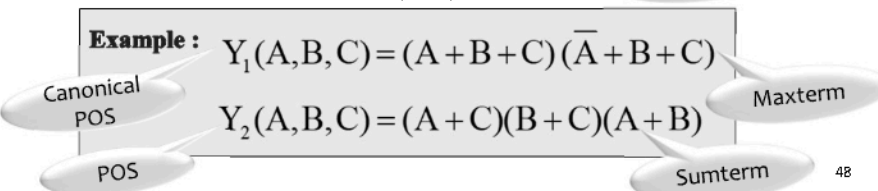
SOP & POS

♦ Any Boolean function can be expressed in two different standard forms.

➤ Sum Of Products (SOP),



➤ Product Of Sums (POS),



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SOP

A	B	C	Y(O/P)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

$(\overline{A}BC)$

$(\overline{A}BC)$

(ABC)

$$Y = (\overline{A}BC) + (\overline{A}BC) + (ABC)$$

$$Y(A, B, C) = \sum m(2, 3, 6)$$

To get Complement, Write only the leftout Terms

$$\overline{Y}(A, B, C) = \sum m(0, 1, 4, 5, 7)$$

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POS

A	B	C	Y(O/P)
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

$(A + \overline{B} + C)$

$(A + \overline{B} + \overline{C})$

$(\overline{A} + \overline{B} + C)$

$$Y = (A + \overline{B} + C) \cdot (A + \overline{B} + \overline{C}) \cdot (\overline{A} + \overline{B} + C)$$

$$Y(A, B, C) = \prod M(2, 3, 6)$$

To get Complement, Write only the left out Terms

$$\overline{Y}(A, B, C) = \prod M(0, 1, 4, 5, 7)$$

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Positive Logic Vs Negative Logic

Positive Logic		Negative Logic	
Voltage Level	Logical Value	Voltage Level	Logical Value
LOW	0	LOW	1
HIGH	1	HIGH	0

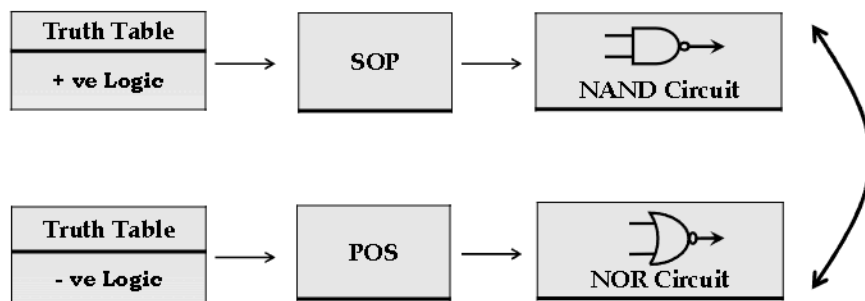
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

A	B	Y
1	1	1
1	0	1
0	1	1
0	0	0

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SOP Vs POS



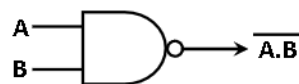
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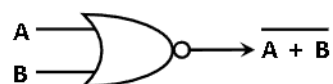
Universal Logic Elements

- ♦ NAND & NOR Gate - Universal Logic Elements
- ♦ Any Boolean Expression can be realized using these gates alone .

NAND Gate



NOR Gate

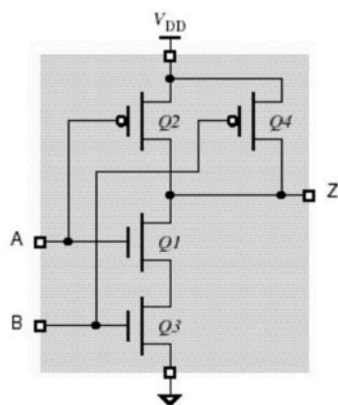


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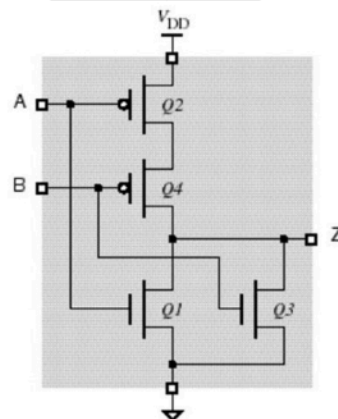
NAND Vs NOR

NAND



NAND - Faster and Smaller

NOR



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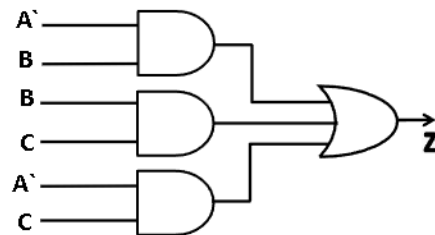
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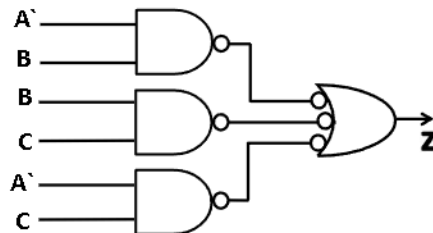
SOP - NAND Gates

Example : $Z = \overline{A}B + BC + \overline{A}C$

Step 1: Implement the Logic Circuit Normally.



Step 2: Replace the intermediate connecting wires with a pair of inverters connected in cascade.



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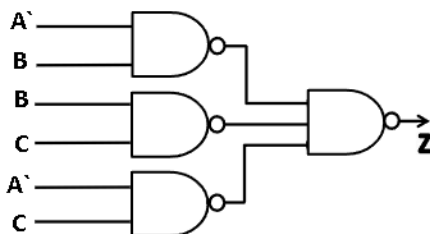
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SOP - NAND Gates

Example : $Z = \overline{A}B + BC + \overline{A}C$

Notice the fact that a pair of inverters in cascade doesn't alter the functionality

Step 3: Replace bubbled OR gate with NAND Gate.



Now, the logic circuit consists of NAND gates only

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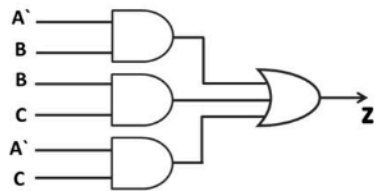
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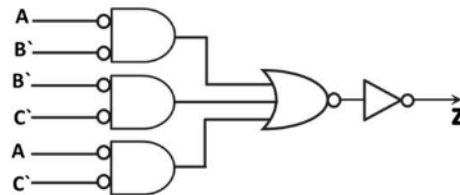
SOP - NOR Gates

Example : $Z = \overline{A}B + BC + \overline{A}C$

Step 1: Implement the Logic Circuit Normally.



Step 2: Replace the Input & Output wires with a pair of inverters connected in cascaded.



Notice the fact that inputs are also toggled to retain the functionality.

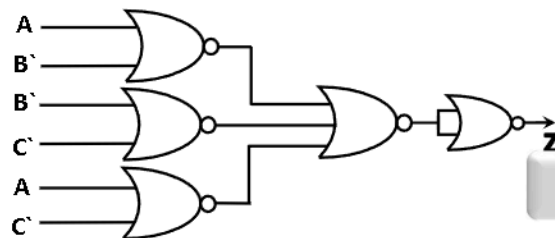
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SOP - NOR Gates

Example : $Z = \overline{A}B + BC + \overline{A}C$

Step 3: replace the bubbled AND Gates at the inputs with NOR Gate & replace the Inverter with 2 i/p NOR Gate with both the i/p's shorted.



Now, the logic circuit consists of NOR gates only

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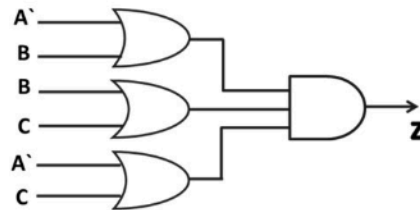
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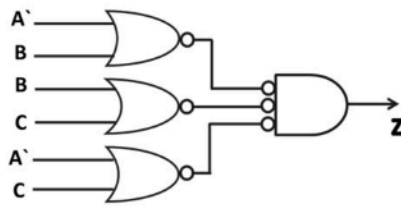
POS - NOR Gates

Example : $Z = (\bar{A} + B).(B + C).(\bar{A} + C)$

Step 1: Implement the Logic Circuit Normally.



Step 2: Replace the intermediate connecting wires with a pair of inverters connected in cascade.



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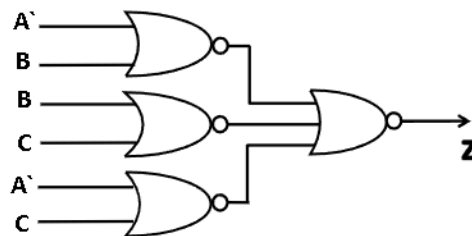
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POS - NOR Gates

Example : $Z = (\bar{A} + B).(B + C).(\bar{A} + C)$

Notice the fact that a pair of inverters in cascade doesn't alter the functionality

Step 3: Replace bubbled AND gate with NOR Gate.



Now, the logic circuit consists of NOR gates only

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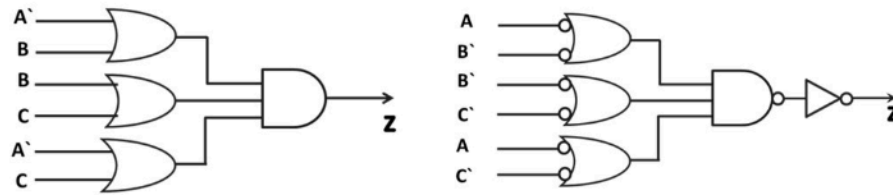
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POS - NAND Gates

Example : $Z = (\bar{A} + B).(B + C).(\bar{A} + C)$

Step 1: Implement the Logic Circuit Normally.

Step 2: Replace the Input & Output wires with a pair of inverters connected in cascaded.



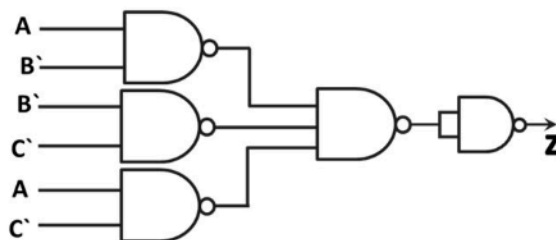
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POS - NAND Gates

Example : $Z = (\bar{A} + B).(B + C).(\bar{A} + C)$

Step 3: replace the bubbled OR Gates at the inputs with NAND Gate & replace the Inverter with 2 i/p NAND Gate with both the i/p's shorted.



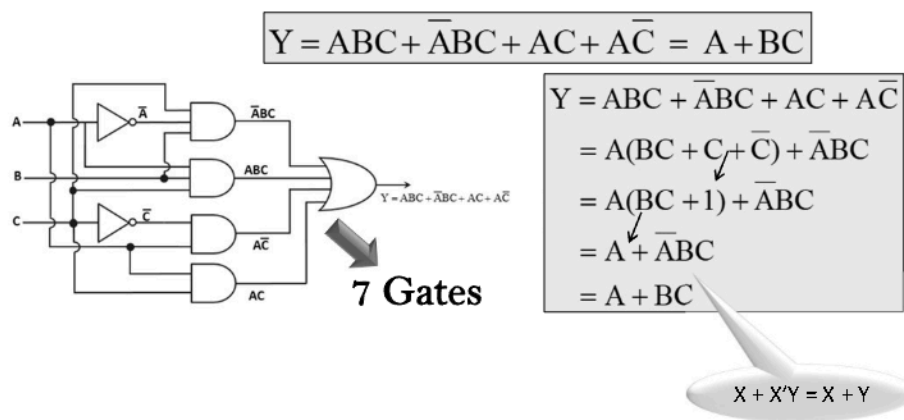
Now, the logic circuit consists of NAND gates only

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Implement the function



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Logic Minimization

Advantages of Logic Minimization

- ♦ Less no. of gates for implementation.
- ♦ Size of the circuit will be less.
- ♦ Less amount of power consumption.
- ♦ More speed of operation of the Logic Circuit.

A major step in digital design

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Logic Minimization

Minimize the following expression

$$Y = \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC\overline{C}$$

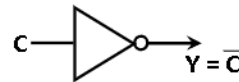
$$Y = \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC\overline{C}$$

$$Y = (\overline{A}\overline{B} + \overline{A}B + A\overline{B} + AB)\overline{C}$$

$$Y = [\overline{A}(\overline{B} + B) + A(\overline{B} + B)]\overline{C}$$

$$Y = (\overline{A} + A)\overline{C}$$

$$Y = \overline{C}$$



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Logic Minimization

Minimize the following expression

$$Y = B\overline{C}\overline{D} + A\overline{B}\overline{C} + A\overline{C}\overline{D} + A\overline{B}\overline{D} + \overline{A}\overline{B}\overline{D}$$

$$Y = B\overline{C}\overline{D} + A\overline{B}\overline{C} + A\overline{C}\overline{D} + A\overline{B}\overline{D} + \overline{A}\overline{B}\overline{D}$$

$$Y = B(\overline{C}\overline{D} + A\overline{C} + A\overline{D}) + A(\overline{B}\overline{C} + \overline{C}\overline{D} + \overline{B}\overline{D})$$

$$Y = B(A\overline{C} + A\overline{D}) + A(\overline{B}\overline{C} + \overline{B}\overline{D})$$

$$Y = A\overline{B}\overline{C} + \overline{A}\overline{B}\overline{D} + A\overline{B}\overline{C} + A\overline{B}\overline{D}$$

$$Y = A\overline{B}\overline{C} + \overline{A}\overline{B}\overline{D} + A\overline{B}\overline{D}$$

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Karnaugh Map

CD AB				
	00	01	11	10
00	0000 0	0001 1	0011 3	0010 2
01	0100 4	0101 5	0111 7	0110 6
11	1100 12	1101 13	1111 15	1110 14
10	1000 8	1001 9	1011 11	1010 10

These cells marked with a specific numbers to represent the corresponding minterm or maxterm

Two adjacent cells are differed by only one digit because of the usage of gray code

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Karnaugh Map

2 Variable K- Map

A \ B	0	1
0	0	0
1	0	1

AB

3 Variable K- Map

A \ BC	00	01	11	10
0	0	1	1	0
1	0	1	1	1

AB + C

4 Variable K- Map

CD AB	00	01	11	10
00	0	1	0	0
01	0	1	0	0
11	1	1	1	1
10	0	1	0	0

AB + C'D

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Karnaugh Map

$$F = \sum m(0, 1, 4, 6, 7)$$

		BC			
A		00	01	11	10
	0	1	1	0	0
	1	1	0	1	1

$$F = AB + A'B' + B'C'$$

		BC			
A		00	01	11	10
	0	1	1	0	0
	1	1	0	1	1

$$F = AB + A'B' + AC'$$

Depending on the grouping,
multiple solutions are possible.

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Karnaugh Map

$$F = \sum m(0, 1, 7, 8, 9, 13, 14, 15)$$

		CD			
AB		00	01	11	10
	00	1	1	0	0
	01	0	0	1	0
	11	0	1	1	1
	10	1	1	0	0

$$F = B'C' + ABC + BCD + ABD$$

		CD			
AB		00	01	11	10
	00	1	1	0	0
	01	0	0	1	0
	11	0	1	1	1
	10	1	1	0	0

$$F = B'C' + ABC + BCD + AC'D$$

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Karnaugh Map - POS

$$F = \prod M(1, 3, 4, 6, 11, 12)$$

CD \ AB	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	1
10	1	1	0	1

$$F = (B' + C + D)(A + B + D')(B + C' + D')(A + B' + D)$$

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Karnaugh Map

$$F = \prod M(1, 3, 4, 6, 11, 12) = \sum m(0, 2, 5, 7, 8, 9, 10, 13, 14, 15)$$

CD \ AB	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	1
10	1	1	0	1

$$F = B'D' + BD + AC'D + ABC$$

CD \ AB	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	1
10	1	1	0	1

$$F = B'D' + BD + AB'C' + ACD'$$

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Don't Care Conditions

- ♦ An input-output conditions that never occurs during normal operation.
- ♦ It is indicated by 'X' in the truth table.
- ♦ It can be either 0 or 1, whichever produces simpler logic circuit.

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Don't Care Conditions

$$F(A, B, C, D) = \sum m(2, 3, 4, 6, 11, 12) + d(10, 13, 14)$$

		CD			
AB		00	01	11	10
		00	01	11	10
00		0	0	1	1
01		1	0	0	1
11		1	X	0	X
10		0	0	1	X

$$F = BD' + B'C$$

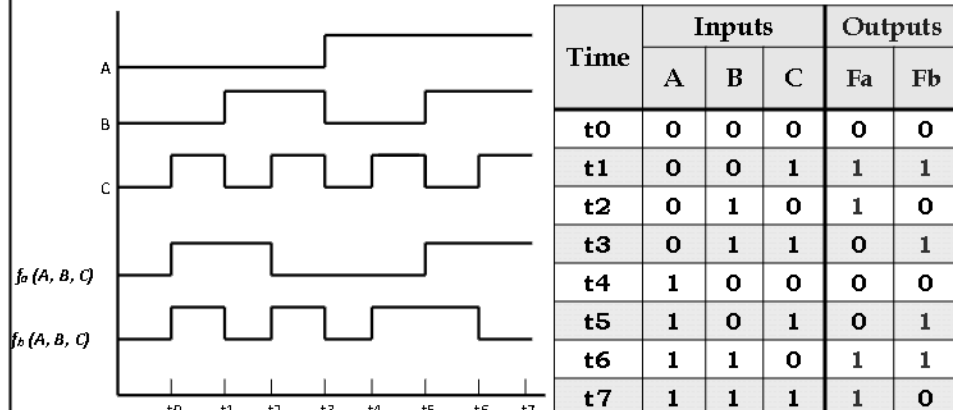
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K Map Example

Design a circuit for the following output waveforms.



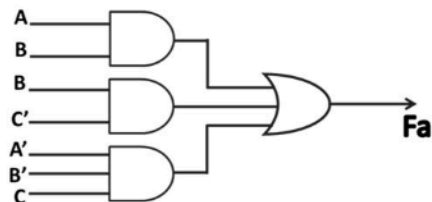
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K Map Example

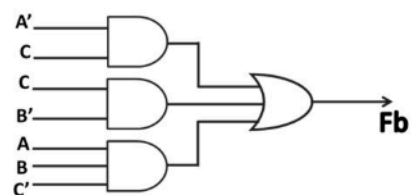
		BC			
A	0	00	01	11	10
	1	0	0	1	1

$$F_a = AB + BC' + A'B'C$$



		BC			
A	0	00	01	11	10
	1	0	1	0	1

$$F_b = A'C + B'C + ABC'$$



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